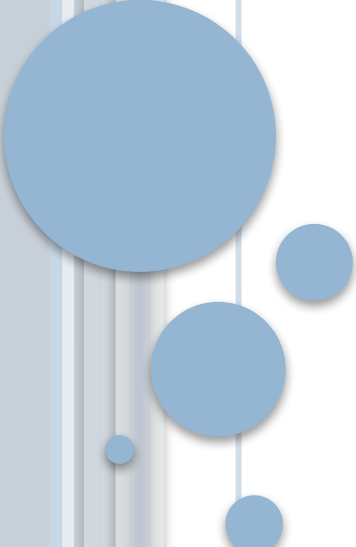




U.2. CRUDE DRUGS

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CRUDE DRUGS

- Crude drugs: naturally occurring, unrefined substance derived from plant, animal, bacteria, organs or whole organisms intended for use in the diagnosis, cure, mitigation, treatment, or prevention of disease in man or other animals.

CRUDE...

- Crude are not advanced in value or improved in or treatment beyond what is essential to its proper packing and the prevention of decay or deterioration pending manufacture.
- Crude Drugs: are not pure compounds.

CRUDE...

- Crude drugs are used infrequently as therapeutic agents;
 - chief principles are separated by various means and are employed in a more specific manner.

CRUDE...

- I Animal Drugs .
 - a. Entire animals
 - Cantharides: an irritant rubefacient
 - **Cochineal or coccus: coloring agent in toothpastes**
 - b. Animal Products
 - Musk: fixative in high-class perfumes
 - Thyroid: in goiter and in obesity
 - Pepsin: assist gastric digestion
 - ii. Vegetable drugs: botanical drugs

NOMENCLATURE OF CRUDE DRUGS

- Common, vernacular or local names
- Scientific names
 - more appropriate for exchange of information
 - they avoid ambiguity

TAXONOMY

- **The binomial system**
- is due to Swedish biologist Linnaeus
- in this system the first name, which is always spelt with Capital letter, denotes the genus
- the second name denotes the species.
- It is however, still equally correct to use capital where the species is named after a person
- the species of Cinchona named after Charles Ledger, who brought its seed from Brazil 1865, is known as *Cinchona Ledgeriana*

TAXONOMY...

- The specific name is usually chosen to indicate:
- 1- Some striking characteristics of the plant
 - *Conium maculatum* (maculate = spotted)
- 2- A characteristic colour:
 - *Veratrum viride* = green
- 3- An aromatic plant or certain aroma:
 - a- *Myritaceae fragrans* (having a fragrant, nice aroma)

TAXONOMY...

- 4- A geographical source or history of a drug:
 - a- *Cannabis indica* =growing in India
- 5- A Pharmaceutical activity or an active constituents:
 - *Papaver somniferum* (sleep inducing
- 6- A general meaning or a special indication
 - a- *Allium sativum* = cultivated

TAXONOMY...

- The generic name may also allude to certain
- characters of the plant:
 - *Glycyrrhiza* is from glucose= sweet, riza = root.

ORIGIN OF CRUDE DRUGS

- The natural or biological origin
- Geographical Sources
- Commercial Origin

ORIGIN...

- **I. The natural or biological origin**
- is the plant or animal yielding it
- if a plant, botanical origin or botanical source
- if an animal, zoological origin or source
- Knowledge of the biological origin enables one to indicate the proper right material and the precise article one which to obtain

ORIGIN...

- **2. Geographical sources of drugs**
- is the region in which the plant or animal grows
- Plants growing in native countries=**indigenous**
e.g. *Aconitum napellus* of the mountainous(Europe)
- *Hyoscyamus muticus*= Egypt
- *Cannabis sativa*=India
- Plants grow that in a foreign land or in locality other than their native home **naturalized**

ORIGIN...

- **3. Commercial origin**

- The commercial origin of a drug refers to its production and its channels of trade
- Alexandrian senna is the product growing in Khartom but it was shipped by the way of Alexandria.

CLASSIFICATION OF CRUDE DRUGS

- Classification is a means of the systematic arrangement of large varieties or quantities of things into smaller groups on the basis of some similar characters present in them.
- A method of classification should be simple, easy to use, and free from confusion and ambiguities.

TYPES OF CLASSIFICATION OF CRUDE DRUGS

- Alphabetic
- Taxonomic
- Morphological
- Pharmacological or therapeutic
- Biogenic/Chemical Classification

I. ALPHABETICAL CLASSIFICATION

- Uses Latin or vernacular names (local names)
- Used in pharmacopoeia
- In European pharmacopoeia, these are arranged according to their names in Latin, where in USP and BPC, these are arranged in English.

ALPHABETICAL...

- Some of the examples are as follows: Acacia, Benzoin, Cinchona, Dill, Ergot, Fennel, Gentian, Hyoscyamus, Ipecacuanha, Jalap, Kurchi, Liquorice, Mints, Nux-vomica, Opium, Podophyllum, Quassia, Rauwolfia, Senna, Vasaka, Wool fat, Yellow bees wax, Zeodary

ALPHABETICAL...

○ Advantage

- It is easy and quick to use;
- no repetition of entries and is devoid of confusion
- location, tracing and addition of drug entries are easy

○ Disadvantage

- no indication of the interrelationships between
plants (drugs)

2. TAXONOMIC CLASSIFICATION

- is purely a botanical classification and is based on the principles of natural relationship and evolutionary developments.
- Thus crude drugs are arranged a in phyla, families, genera and species, using one of the accepted systems of the botanical classification.

TAXONOMIC...

○ Taxonomical classification of some crude drugs

Phylum	Order	Family	Drugs
Thallophyta (Algae, Fungi, Lichens)	Gelidiales Hypocreales Lecanorales	Gelidiaceae Clavicipitaceae Parmeliaceae	Agar ergot Usnea spp.
Bryo- and Pteridophyta (Liverhorts, Mosses and Ferns)	Marchantiales Sphagnales Dryopteridales	Marchantiaceae Sphagnaceae Dryopteridaceae	Marchantia Sphagnum Dryopteris/ male fern

TAXONOMIC....

- **Advantage:**

- Precise & ordered arrangement (no ambiguity)

- **Disadvantage:**

- does not correlate in between the chemical constituents and biological activity of the drugs
- fails to recognize the organized and unorganized nature of crude drugs and chemical nature of active constituents and therapeutic significance of crude drugs

3. MORPHOLOGICAL CLASSIFICATIONS

- Arranged according to the part or organ of the plant and animal used
- Drugs are divided into
- **Organized:**
 - direct parts containing cellular tissues
- **Unorganized drugs:**
 - prepared by incision, drying, or extraction with a solvent
 - not containing any cellular plant or animal tissues

MORPHOLOGICAL CLASSIFICATION

○ **Organized drugs**

- Leaves
- Flowers
- Fruits
- Seeds
- Whole organisms
- Rhizomes
- Woods
- Barks
- Herbs
- Roots

○ **Unorganized drugs**

- Resins
- Oils
- Dried latices
- Extracts
- Fats
- Waxes
- Gums

MORPHOLOGICAL...

Classification of drugs	Drug examples
Organized drugs	
Barks	Arjuna, Ashoka, Cascara, Cassia, Cinchona, Cinnamon. Kurchi, Quillaia, Wild Cherry
Flowering parts	Clove, Pyrethrum, Saffron, Santonica, Chamomile, Artemisia
leaves	Digitalis, Eucalyptus, Gurniar, Pudina, Senna, Spearmint. Squill, Tulsi, Vasaka, Coca, Buchu, Hamamelis, Hyoscyamus, Belladonna, Tea

MORPHOLOGICAL....

○ Advantage

- more helpful to identify and detect adulteration
- more convenient for practical study, especially when the chemical nature of the drug is not clearly understood.

○ Disadvantage

- Microscopical studies is needed (powdered herbs)
- Repetition of drugs or plants occurs

4. PHARMACOLOGICAL/THERAPEUTIC CLASSIFICATION

- Plant drugs are grouped according to their pharmacological action of the most important active constituent in the plant

PHARMACOLOGICAL...

- Classification of drugs based on pharmacological action

Pharmacological or therapeutic classes	Name of drugs
--	---------------

Malignant growth inhibitors	
------------------------------------	--

1. Anticancer (suppresses malignant growth)	1. Vinca, Podophyllum, Taxus, Camptotheca
---	---

2. Anti-inflammatory (suppresses inflammation)	2. Colchicum corm and seed, turmeric
--	--------------------------------------

PHARMACOLOGICAL...

○ Advantages

- This system of classification can be used for suggesting substitutes of drugs if they are not available at a particular place or time.

○ Disadvantage:

- The constituents of one drug may have more than one therapeutic action (fall into numerous groups)
- e.g. Flavonoids

5. BIOGENIC/CHEMICAL CLASSIFICATION

- Drugs are classified according to the main active chemical constituent available in the plant
- More than 75 pure compounds derived from higher plants find their place in modern medicine.
- For example
 - codeine, atropine, ephedrine, hyoscyamine, digoxin, hyoscine, digitoxin, pilocarpine, theobromine, theophylline, quinidine, quinine, emetine, caffeine, papaverine, and colchicine

CHEMICAL...

- Classification of drugs on the basis of chemical nature

Chemical constituents	Drugs	Plant sources
1.	Carbohydrates (i) Gum (ii) Mucilage (iii) Others	(i) Acacia, Tragacanth, Guar gum, Sterculia (ii) Plantago seed (iii) Starch. Honey, Agar,
2.	Glycosides (i) Anthraquinone (ii) Saponins	(i) Aloe, Cascara, Rhubarb, Senna (ii) Quillaia, Arjuna, Glycyrrhiza, Dioscorea
3.	Volatile oils	Cinnamon, Coriander, Cardamom, Orange peel, Mint, Clove, Ginger,

CHEMICAL....

○ Advantage

- Popular for teaching since biological activities of crude drugs (therapeutic and pharmacological activities) are based on the chemical constituents

○ Disadvantage

- Ambiguities
 - Plants contain more than one group of active constituents each with different actions

EVALUATION OF CRUDE DRUGS METHODS

- It deals with the correct identification of the plant, determination of the quality & purity & detection of nature of adulteration of the crude drugs.
- Reasons for crude drug evaluation:
 - biochemical variation in the drug
 - deterioration due to treatment & storage
 - substitution and adulteration

EVALUATION...

- **Identification of the plant**
- The **identity** of a drug can be established by:
 - Actual collection of the drug from the plant or animal that has been previously identified
 - Another method of identification is by comparing a representative unknown sample with a published description of the drug with authentic drug samples.

QUALITY MAINTAINING METHODS

- The most important means of accomplishing this are:
 - Collection of the plant from the correct natural source at the proper time
 - Preparation by proper cleaning, drying and garbling
 - Proper preservation against contamination

EVALUATION...

- The evaluation of the drug involves a number of methods, which may be classified as follows:
 - 1. Organoleptic (Morphological) Evaluation
 - 2. Microscopic Evaluation
 - 3. Chemical Evaluation
 - 4. Physical Evaluation
 - 5. Biological () Bioassay) Evaluation

I. ORGANOLEPTIC (MORPHOLOGICAL) EVALUATION

- evaluation by means of the organs of sense and includes the macroscopic appearance of the drug
 - Shape and size.
 - Colour and external markings
 - Fracture and internal colour
 - Odour and taste

2. MICROSCOPIC EVALUATION

- essential for the study of the adulterants in powder and in the identification of the pure powdered drug.
 - The leaf constants like:
 - stomatal number
 - stomatal index
 - palisade ratio
 - vein islet
 - vein termination number

3. CHEMICAL EVALUATION

- It comprises of different chemical test & chemical assays
- The isolation, purification & identification of active constituents are chemical methods of evaluation.
 - Phytochemical screening

CHEMICAL IDENTIFICATION TESTS

- Alkaloids
 - Dragendorff's reagent
- Phenolic compound & tannins
 - Ferric chloride test
- Anthraquinone glycoside
 - Borntrager's test
- Saponin glycoside
 - Foam test
- Flavonoids glycoside
 - Lead acetate test
- Fixed Oils & Fats
 - Spot Test
- Carbohydrates
 - Molisch's Test

4. PHYSICAL EVALUATION

- Moisture contents
- Melting point,
- boiling point,
- refractive index,
- ash value,
- extractive values,
- chromatographic techniques

5. BIOLOGICAL EVALUATION (BIOASSAY)

- Biological evaluation includes the quantitative assay of the drugs using biological models of intact animals, animal preparations, isolated living tissues or microorganisms

BIOLOGICAL...

- **Generally used if: -**
- The chemical identity of the active principle is not fully elucidated
- If no adequate chemical assay has been derived for the active principle as in case of insulin
- If the chemical assay is not a valid indication of biological activity

BIOLOGICAL...

- If the drug is composed of complex mixture and activity. E.g. Digitalis,
- If purification of the crude drug is not possible E.g. Separation of vit. D from irradiated oils
- It measures the actual biological activity of a given sample

BIOLOGICAL...

- The disadvantages are: -
- Quantitative accuracy is usually less than observed with most chemical analysis
- Techniques & interpretations involved vary with different operators
- The effect measured in the test animals is different form that observed in treating patients.

BIOLOGICAL...

○ *In-Vitro* Bioassays

- Rapid
- less selective
- Diagnostic
- less expensive
- suitable large number of

samples.

○ *In-Vivo* Bioassays

- Selective
- expensive
- Time Consuming
- Animal model
- require large amounts of material

OFFICIAL AND UNOFFICIAL DRUGS

- **Pharmacopoeias:** book recognized by the government as the legal authority for standardization of drugs.
- The official preparations are standardized; obtained of uniform strength throughout the country.
 - Are listed and described as being a definite therapeutic agent in the Pharmacopoeia.

MONOGRAPH OF A CRUDE DRUG

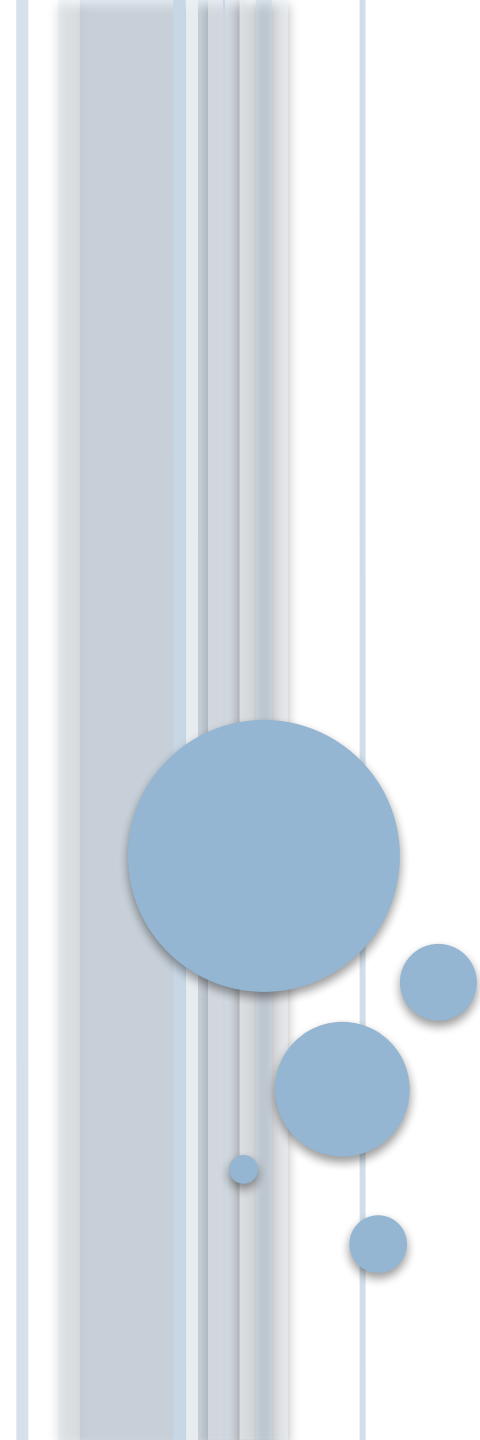
- English for the market
- Arabic
- Latin
- French names
- Definition
- Description
- Special condition for collection or preparation
- Identity tests
- Tests for adulterant
- Method of assay,
- Special storage requirements
- the usual dose and range of dosage if the drug is administered internally

TYPES OF PREPARATIONS FROM PLANTS

- Fresh plant
- Crude drugs
- Crude extracts (extractives)
- Fractions
 - Solvent fractions
 - Class of natural products.
E.g. Flavonoidal, alkaloidal, saponin fraction etc.
- Isolated pure compound
E.g. Reserpine from *Raulfia serpentina*

PARAMETERS TO CONSIDER

- The decision at which stage to make the drug available to consumers depends on:
 - Safety factors
 - Efficacy factors
 - Quality
 - Economical factors



U.3. STEPS IN THE SCIENTIFIC ANALYSIS OF DRUGS FROM NATURAL RESOURCES

3.1. INTRODUCTIONS

- Are processes required for finding the most effective active principles starting from crude drugs.
- Should be followed and documented thoroughly to decrease errors and facilitate repetition if necessary

I. SELECTION AND COLLECTION OF PLANT MATERIAL

- Accidental experience
- Doctrine of signatures
 - The notion that nature has put a clue (signature) for the medicinal value of plant that is the association of certain features of a plant with a certain disease

SELECTION...

- Examples:
- *Plantago lanceolata* has leaves that look like a foot and so was used by caravan merchants for sore foot and foot wounds.
- *Hepatica acutiloba* (liverwort), has a three-lobed leaf that resembles liver and used in treating liver ailments.
- *Raulfia serpentina* has roots that looks like snake (serpent) and was used for sneak bite.

○ *Plantago lanceolata*



Raulfia serpentina (roots)



SELECTION...

- Random screening
- Traditional use
 - Less expensive and more efficient
 - Ethnobotanical selection of medicinal plants
 - Recommended by WHO
- Literature Information Selection Technique (L.I.S.T.)
 - NAPRALERT® database.

COLLECTION

- The collection of samples could be performed from
 - Wild: forest, natural habitat
 - Cultivated: botanical gardens, research institutions
- The collection could also be performed by
 - Skilled man power
 - Labor force

GULLELE BOTANIC GARDEN

- It started (2012) with 250 plant species, which have now increased to 700.



2. TAXONOMIC IDENTIFICATION OF THE PLANT

- Establishment of the exact identity of the plant
(kingdom, subkingdom, division, class, order family, genus, species, variety)
- Experienced field botanist is usually required.
- Helps to avoid mistakes over plant identity & for further reference

IDENTIFICATION...

- Pressing method is used to pack plant materials
- Pressing method is a means of wrapping the desired plant in a file envelop by applying force to fix the plant on to the envelop

PRESSING AT HERBARIUMS



3. LITERATURE SURVEY ON THE IDENTIFIED PLANT

- Helps to avoid repetition and get update on current knowledge about the plant.
- It also helps to check consistency of folkloric use
- Primary data, journals (ethnobotanical, ethnopharmacological) are used to gather information
- Secondary data for gather information include
 - PubMed central
 - Hinary

PubMed central web page

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4. DESIGN OF APPROPRIATE EXTRACTION METHOD

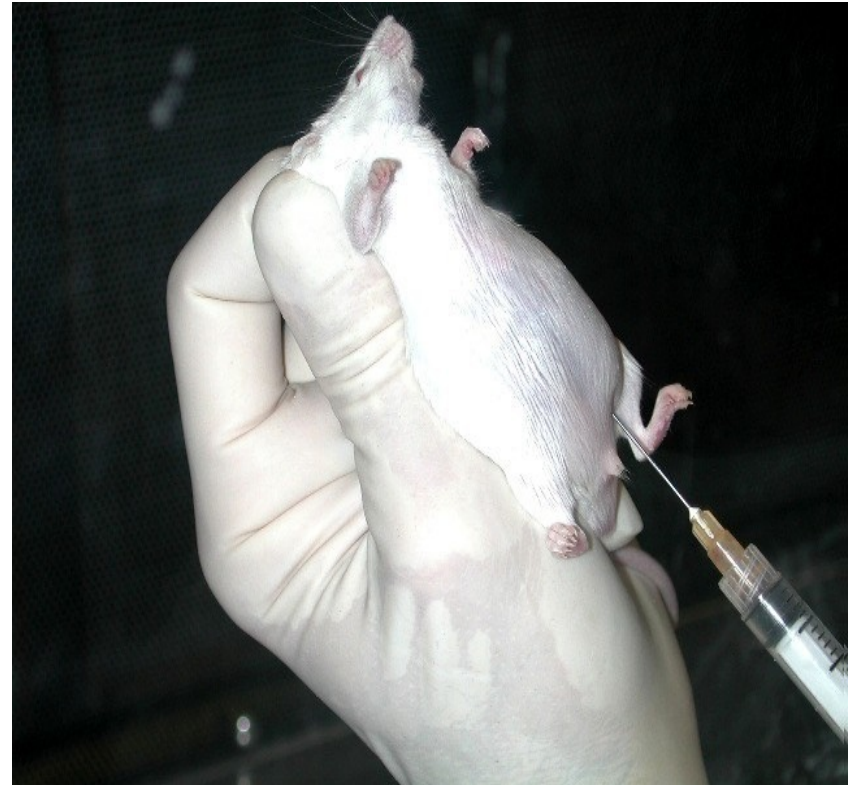
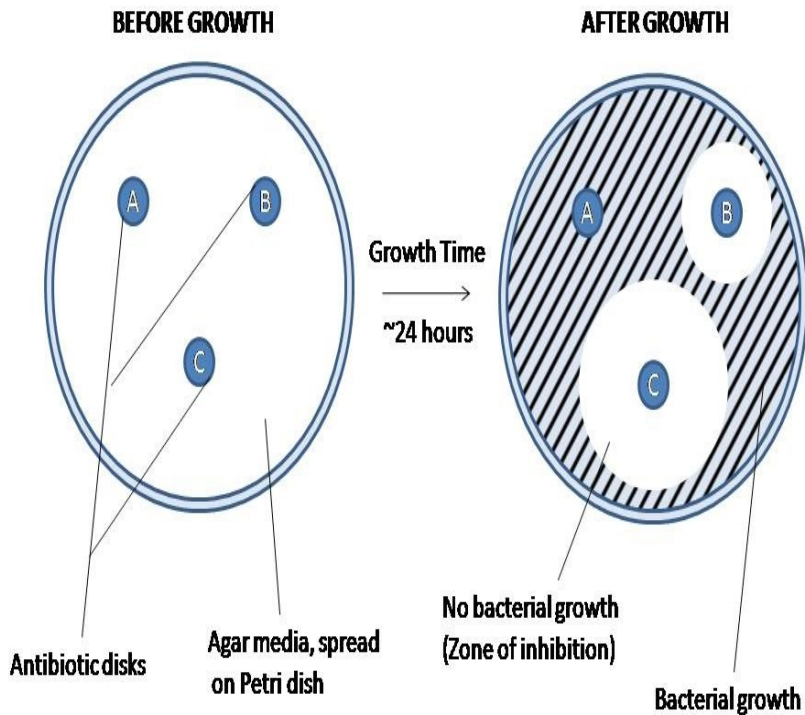
- Should mimic the extraction methods used in traditional medicine to target active principles
- Could also be obtain from previous works conducted on similar plants or plants belonging to similar genus
- Should give high yields to gain a number of potential candidate (active principles)

5. CHECKING EXTRACTS/ FRACTIONS FOR PHARMACOLOGICAL ACTIVITY

- Could be done through
 - *In-vitro* and *In-vivo*
 - E.g. Antibacterial tests
 - Disk diffusion method (*In-vitro*)
 - Animal model (mice or rats) (*in-vivo*)

Disk diffusion method (*In-vitro*)

mice (*in vivo*)



6. PHYTOCHEMICAL SCREENING

- Could be
- Chemical tests
 - Identifying the general class of compound
- Chromatography
 - Identifying the number of compounds found in extract or fractions

7. ISOLATION OF ACTIVE COMPOUNDS

- Separating the active principle
- Using
 - Classical: distillation, crystallization, filtration etc
 - modern methods:
 - chromatographic methods like CC, TLC, HPLC, HPTLC

7. STRUCTURE DETERMINATION

- Proposing and confirmation of the structure and chemical formula of the active principle
- By means of
 - Spectroscopic techniques
 - NMR, UV, IR--- arrangement of elements and functional groups
 - MS----- molecular mass

3.2. APPROACHES TO SCREENING PROCESS

○ 3.2.1. Phytochemical approach

- Firstly requires the isolation of a pure compound from the plant
- Secondly, determining its bioactivity by applying the classic pharmacological studies.

3.2. APPROACHES...

- It involves:
 - Screening of extracts for specific chemical class.
 - Using chemical reagents for detection.
 - Chemically-guided fractionation.
- Results:
 - Minute/Trace components are mostly missed (detection limits).
 - Unknown natural compounds are certainly missed.
 - Mostly: No biological activity found

3.2.2. BIOACTIVITY APPROACH

- It involves:
- Extracts are screened for a specific therapeutic activity using mostly *in-vitro* bioassay(s).
- Only those possessing a bioactivity are fractionated
- Active fractions are further fractionated to isolate the active compounds in pure form
- Followed by characterization of its structure

3.2.2. BIOACTIVITY.....

- Results/often leads to the finding of
 - highly potent,
 - minor,
 - novel compounds

